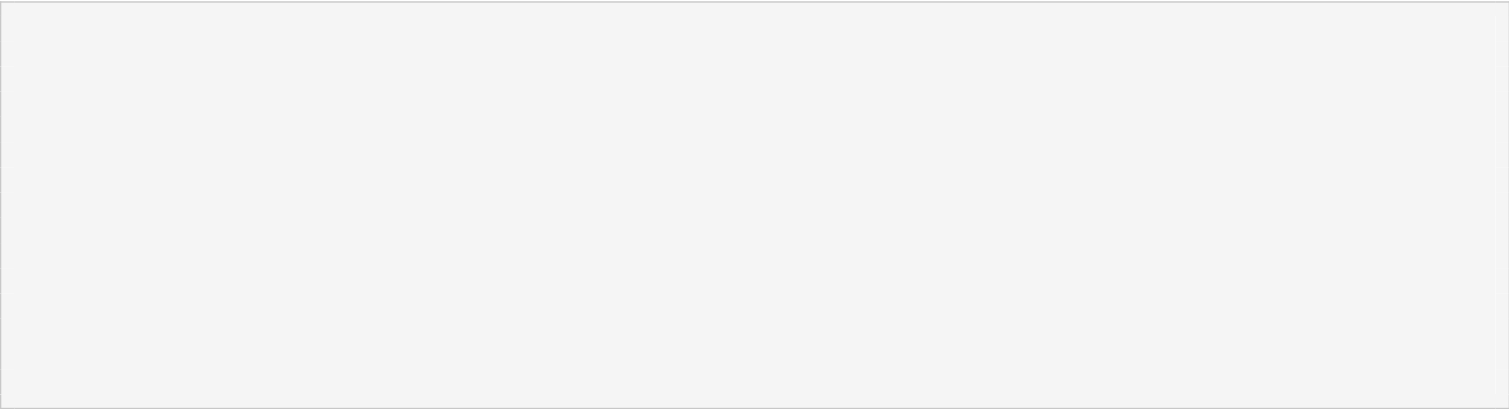
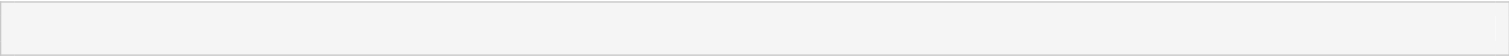


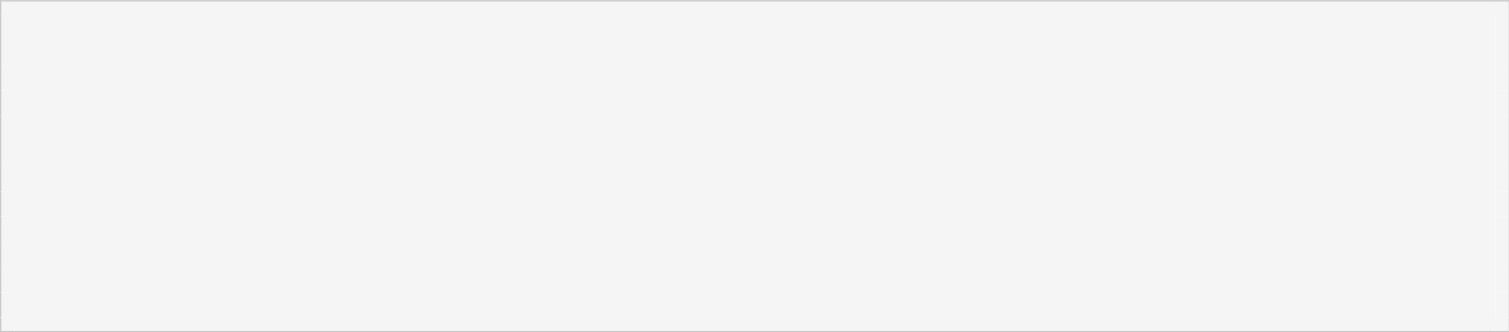
$$\frac{1}{4}$$

$$\frac{1}{4}$$

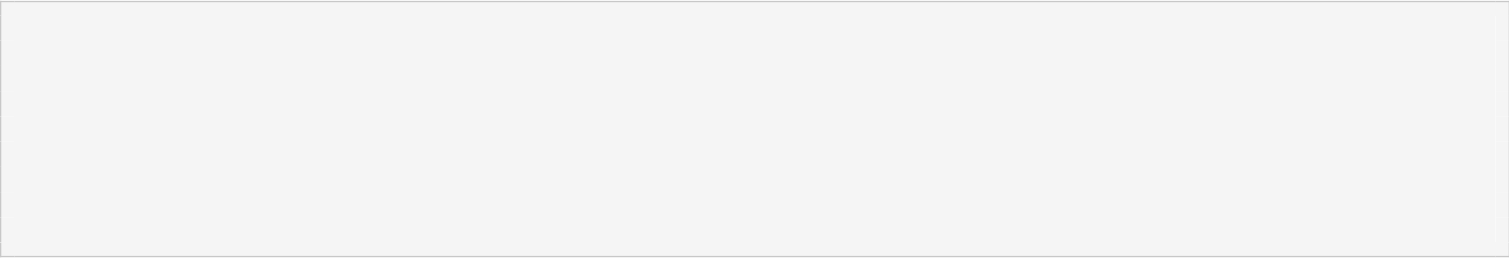
$$(2, 1, 1, 0)$$



$$\pm 2\hat{e}_i 2\sqrt{2}\frac{1}{2}$$



$$(\pm 1, \pm 1, \pm 1)\frac{1}{2}$$



$$(k, k, k, k)$$

$$\|/4\\(2,1,1,0)$$

$$(0,0,0,0)(1,0,0,0)(0,1,0,0)(0,0,1,0)\frac{1}{4}$$

$$\frac{P_0,\ldots,P_3}{VS3=\sqrt{9/8}}$$

$$\mid_{5\times 5}\mid_{3\times 3}\qquad\qquad\qquad\mid\mid/4$$

$$\frac{1}{4}$$

$$(P_1-P_0)$$

$$d_{ij}P_iP_j$$

$$V_{ivm}=S3\cdot V_{xyz}S3=\sqrt{9/8}$$

$$V_{ivm}=S3\cdot V_{xyz}S3=\sqrt{9/8}$$

$$P_1-P_0P_2-P_0P_3-P_0\frac{1}{4}$$

$$[i,j]=\mathbb{E}[\partial_{\theta_{\boldsymbol{i}}}p(x;\theta)\partial_{\theta_{\boldsymbol{j}}}p(x;\theta)]$$

$$_{288}V^2=\begin{pmatrix}0&1&1&1&1\\1&0&d_{01}^2&d_{02}^2&d_{03}^2\\1&d_{10}^2&0&d_{12}^2&d_{13}^2\\1&d_{20}^2&d_{21}^2&0&d_{23}^2\\1&d_{30}^2&d_{31}^2&d_{32}^2&0\end{pmatrix}$$

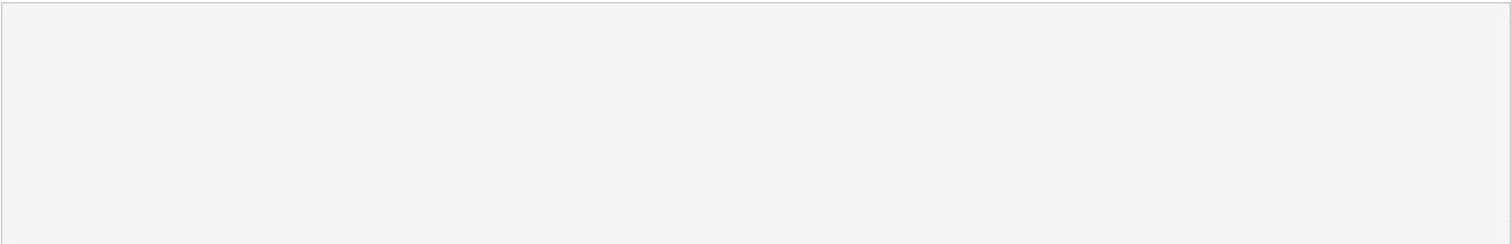
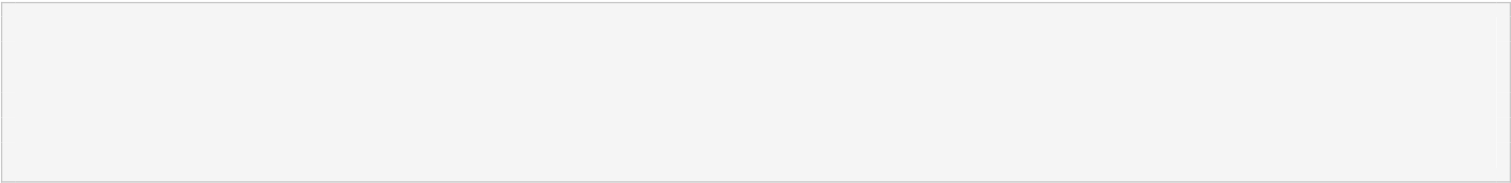
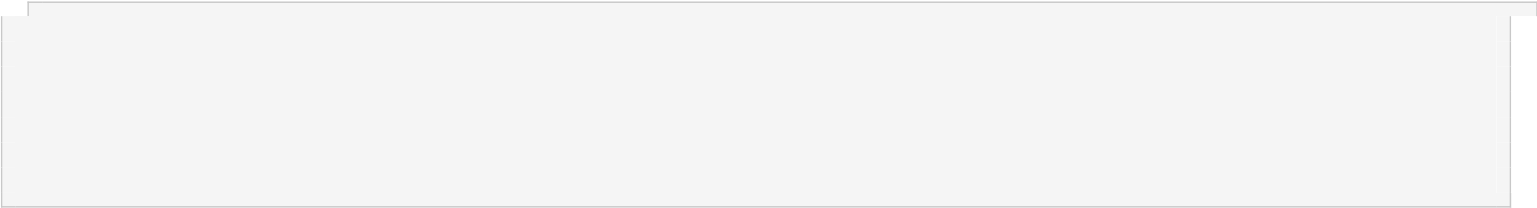
$$S3=\sqrt{9/8}$$

$$a,b,cd,e,f$$

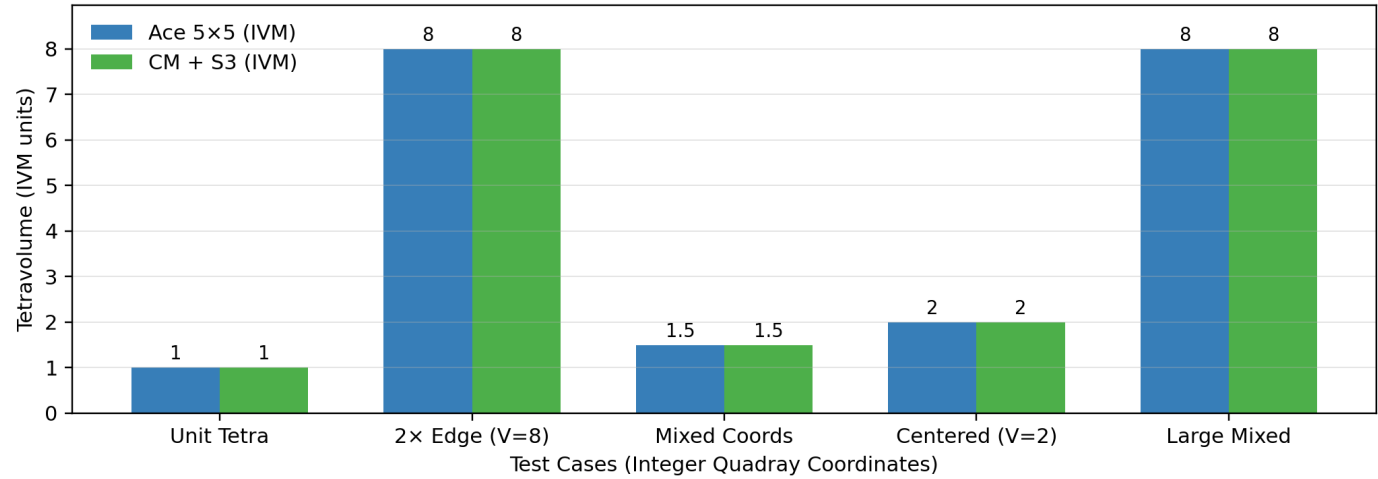
$$V_{ivm}=S3\cdot V_{xyz}S3=\sqrt{9/8}$$

$$P_1-P_0P_2-P_0P_3-P_0\frac{1}{4}$$

$$S3=\sqrt{9/8}$$



Validation: Bridging vs Native Tetravolume Formulations

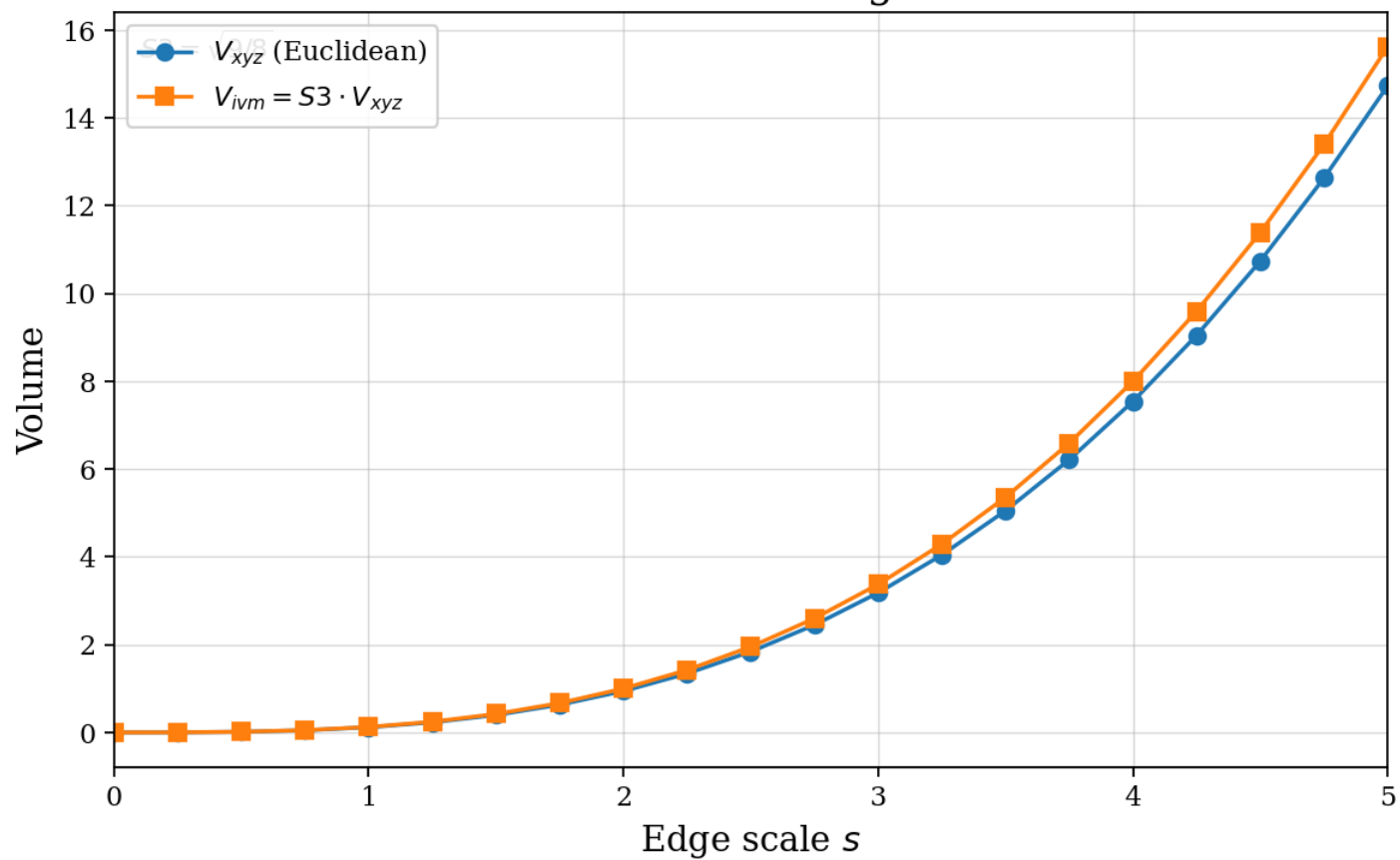


Test cases: Unit tetra (V=1), 2x edge scaling (V=8), mixed coordinates, centered (V=2), large mixed.
Both methods agree exactly (symbolic identity), validating the S3 conversion factor. Data: quadmath/output/data/bridging_vs_native.csv

$S3 = \sqrt{9/8}$

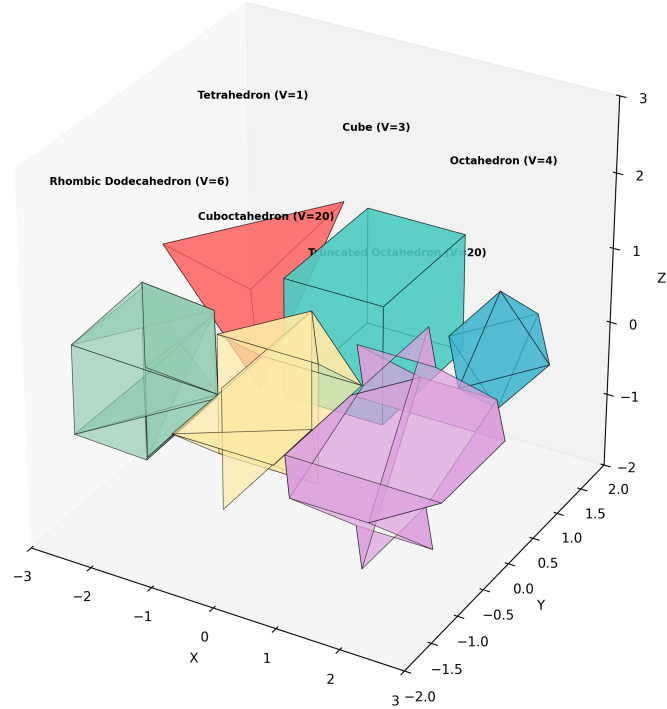
(2, 1, 1, 0)

Tetra volume vs edge scale

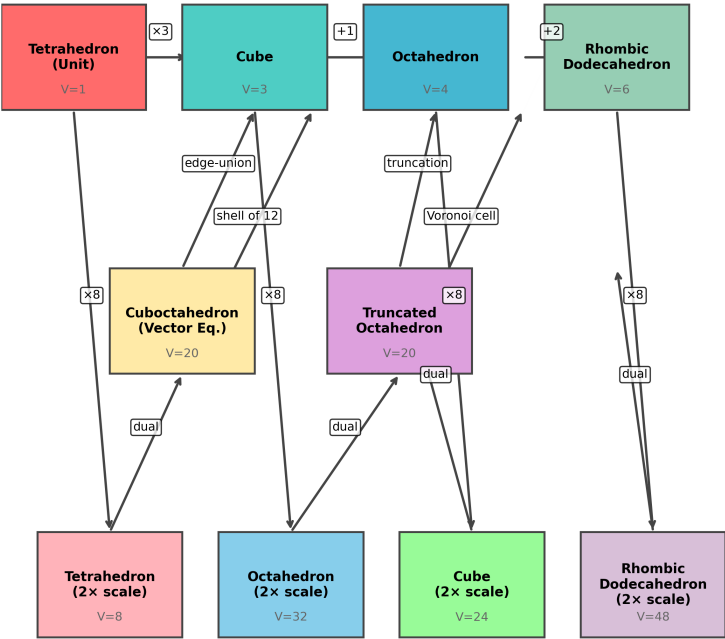


$$V = \frac{\sqrt{2}}{12} \cdot L^3 V_{ivm} = \frac{1}{8} \cdot L^3 S3 = \sqrt{9/8} \approx 1.06066$$

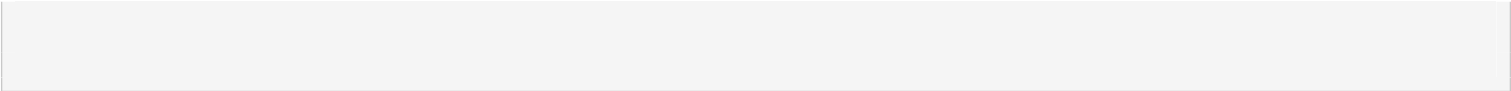
3D Polyhedra Visualizations
(IVM Volume Units)



Synergetic Polyhedra Volume Relationships
Quadray/IVM Framework Network Diagram



Volume relationships in IVM tetra-units. Regular tetrahedron (V=1) is the fundamental unit.
Scaling relationships: $2 \times \text{edge length} \rightarrow 8 \times \text{volume}$ ($V \propto L^3$).
Geometric constructions: edge-union, truncation, dual relationships, and Voronoi cells.
All polyhedra constructed with consistent edge lengths and integer Quadray coordinates.



$A, B, CABC$
 $(P_i - P_0)$



$3 \times 3 \mid \mid /4$



G

